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of Transportation
**Federal Railroad
Administration**

Office of Research,
Development and Technology
Washington, DC 20590

Rail Crossing Violation Warning Concept of Operations



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14. ABSTRACT The Rail Crossing Violation Warning (RCVW) application provides the means for equipped connected vehicles (CV) on approach to a highway-rail intersection (HRI) to be warned of an imminent violation of an active HRI through the interconnection of a CV Roadside-Based Subsystem (RBS) with track-circuit based train detection systems in place at protected HRIs. The system includes a cloud-based subsystem (CBS) which broadcasts HRI information over the internet such as: HRI identification, HRI attributes, train presence and RBS diagnostics to authorized subscribers. This document describes the Concept of Operations (ConOps) for the Vehicle-to-Everything (V2X) Rail Crossing Violation Warning (RCVW). This document supersedes and replaces the RCVW Concept of Operations documents created under FRA contract number 693JJ6-18C-000019.					
15. SUBJECT TERMS Rail Crossing Violation Warning, RCVW, connected vehicles, CV, Connected Automated Vehicles, CAV, Highway-Rail Intersection, HRI Dedicated Short Range Communications, DSRC, signal phase and timing, SPaT, Map Data, MAP, Institute of Electrical and Electronics Engineers, IEEE, C-V2X, Cloud-Based Subsystem, CBS					
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METRIC/ENGLISH CONVERSION FACTORS

ENGLISH TO METRIC

LENGTH (APPROXIMATE)

1 inch (in)	=	2.5 centimeters (cm)
1 foot (ft)	=	30 centimeters (cm)
1 yard (yd)	=	0.9 meter (m)
1 mile (mi)	=	1.6 kilometers (km)

AREA (APPROXIMATE)

1 square inch (sq in, in ²)	=	6.5 square centimeters (cm ²)
1 square foot (sq ft, ft ²)	=	0.09 square meter (m ²)
1 square yard (sq yd, yd ²)	=	0.8 square meter (m ²)
1 square mile (sq mi, mi ²)	=	2.6 square kilometers (km ²)
1 acre = 0.4 hectare (he)	=	4,000 square meters (m ²)

MASS - WEIGHT (APPROXIMATE)

1 ounce (oz)	=	28 grams (gm)
1 pound (lb)	=	0.45 kilogram (kg)
1 short ton = 2,000 pounds (lb)	=	0.9 tonne (t)

VOLUME (APPROXIMATE)

1 teaspoon (tsp)	=	5 milliliters (ml)
1 tablespoon (tbsp)	=	15 milliliters (ml)
1 fluid ounce (fl oz)	=	30 milliliters (ml)
1 cup (c)	=	0.24 liter (l)
1 pint (pt)	=	0.47 liter (l)
1 quart (qt)	=	0.96 liter (l)
1 gallon (gal)	=	3.8 liters (l)
1 cubic foot (cu ft, ft ³)	=	0.03 cubic meter (m ³)
1 cubic yard (cu yd, yd ³)	=	0.76 cubic meter (m ³)

TEMPERATURE (EXACT)

$$[(x-32)(5/9)]^{\circ}\text{F} = y^{\circ}\text{C}$$

METRIC TO ENGLISH

LENGTH (APPROXIMATE)

1 millimeter (mm)	=	0.04 inch (in)
1 centimeter (cm)	=	0.4 inch (in)
1 meter (m)	=	3.3 feet (ft)
1 meter (m)	=	1.1 yards (yd)
1 kilometer (km)	=	0.6 mile (mi)

AREA (APPROXIMATE)

1 square centimeter (cm ²)	=	0.16 square inch (sq in, in ²)
1 square meter (m ²)	=	1.2 square yards (sq yd, yd ²)
1 square kilometer (km ²)	=	0.4 square mile (sq mi, mi ²)
10,000 square meters (m ²)	=	1 hectare (ha) = 2.5 acres

MASS - WEIGHT (APPROXIMATE)

1 gram (gm)	=	0.036 ounce (oz)
1 kilogram (kg)	=	2.2 pounds (lb)
1 tonne (t)	=	1,000 kilograms (kg)
	=	1.1 short tons

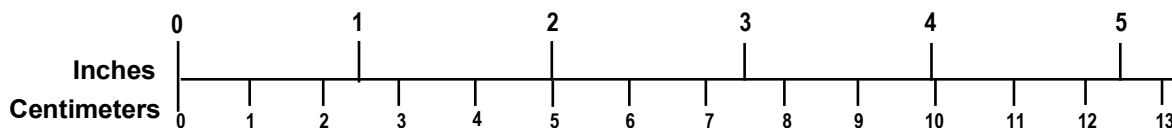
VOLUME (APPROXIMATE)

1 milliliter (ml)	=	0.03 fluid ounce (fl oz)
1 liter (l)	=	2.1 pints (pt)
1 liter (l)	=	1.06 quarts (qt)
1 liter (l)	=	0.26 gallon (gal)
1 cubic meter (m ³)	=	36 cubic feet (cu ft, ft ³)
1 cubic meter (m ³)	=	1.3 cubic yards (cu yd, yd ³)

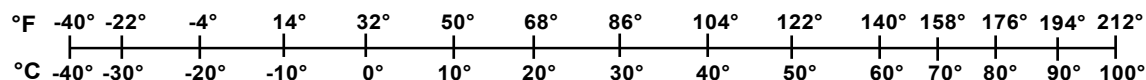
TEMPERATURE (EXACT)

$$[(9/5)y + 32]^{\circ}\text{C} = x^{\circ}\text{F}$$

QUICK INCH - CENTIMETER LENGTH CONVERSION



QUICK FAHRENHEIT - CELSIUS TEMPERATURE CONVERSION



For more exact and or other conversion factors, see NIST Miscellaneous Publication 286, Units of Weights and Measures. Price \$2.50 SD Catalog No. C13 10286

Updated 6/17/98

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Revision History

Revision	Date	Change Description	Affected Chapters/Pages
A	6/17/2019	Initial Release	All
B	9/30/2019	Revised to address FRA and Volpe comments	Various
C	12/07/2022	Addition of a Cloud Based Subsystem Updates to Standards General Edits	Various
D	2/15/2023	Addressed comments by FRA	Various
E	5/5/2023	Addressed additional comments by FRA	Various
F	5/19/2023	Removal of CBS need 013e. Continuously verify CBS Subscriber(s) link communications	Section 4.5

1. Scope

This document describes the Concept of Operations (ConOps) for a Rail Crossing Violation Warning (RCVW) Intelligent Transportation Systems (ITS) Connected Vehicle (CV) safety system. This ConOps describes the current state of operations of the RCVW system, establishes the rationale for including a cloud-based component, and describes the RCVW system in terms of its features and operations.

Prior to the RCVW system concept development, the Cooperative Intersection Collision Avoidance Systems (CICAS) initiative formed a working group by partnering with automotive Original Equipment Manufacturers (OEMs), State and local Departments of Transportation (DOT)s, and university research centers throughout the country. The goals of that initiative were to:

- Develop and demonstrate CICAS for violations of traffic control devices and to assist a CV maneuver through cross traffic
- Assess the value and acceptance of CICAS
- Develop and provide tools to support industry developments

Subsequent to the conclusion of CICAS program, the Federal Railroad Administration (FRA) initiated a project carried out by the Volpe National Transportation Systems Center to leverage the CICAS research to develop a ConOps and System Requirements Specification (SRS) for a conceptual rail collision avoidance system utilizing CV technology. Following the completion of these efforts, the FRA and the ITS – Joint Program Office (JPO) utilized the ConOps and System Requirements Specification (SRS) as the basis for developing a prototype RCVW system.¹ The RCVW system proved that CV technologies could be employed to provide on-board rail crossing violation warnings. It demonstrated that actionable information could be communicated wirelessly from infrastructure to, and for the use by, an approaching CV.

The FRA proceeded to fund Phase II of the RCVW system, taking it from a proof of concept to a reliable, accurate, and widely applicable system. Phase II enhanced the system reliability and incorporated additional functionality that demonstrated the potential for leveraging real-time CV concepts and services to enhance rail crossing safety.²

Leveraging the FRA’s existing investment in RCVW technology, the proposed system described in this ConOps will:

1. expand, enhance, and refine the capabilities of the existing system by including a cloud-based component;
2. incorporate Cellular Vehicle-to-Everything (C-V2X) communications; and
3. identify scenarios in which the cloud-based component might be used to provide benefits of the RCVW concept to roadway vehicles that do not have CV technology.

¹ FHWA-JPO-18-609 “Vehicle-to-Infrastructure Prototype Rail Crossing Violation Warning Application. Project Report

² DOT FRA ORD-22/07 “Rail Crossing Violation Warning Application – Phase II

1.1 Identification

This document represents the first of a series of System Engineering documents being developed for the RCVW system. This document describes the concept of operations for the RCVW system and components. It focuses on describing the current state of practice as well as reintroducing the concepts for the RCVW system. This ConOps is restricted to describing the expected functionality and operation of the RCVW system and its components.

This ConOps supersedes and replaces the RCVW Concept of Operations documents created under FRA contract number 693JJ6-18C-000019.

1.2 Document Overview

This ConOps describes the operational characteristics for a Vehicle-to-Everything (V2X) rail crossing safety application enhanced with a cloud-based component. The concepts presented in this document are intended to improve safety at railroad grade crossings, herein referred to as Highway Rail Intersections (HRI), particularly for vehicles with a planned/predicted approach to an HRI. This ConOps will be revised periodically to reflect lessons learned and additional user needs when they are identified. The United States Department of Transportation (US DOT) FRA is responsible for the configuration control of this document.

This ConOps will:

- Communicate an understanding of how the RCVW system will be designed
- Serve as the foundation from which system requirements are derived
- Serve as a means for informing potential stakeholders

The intended audience of this ConOps includes application developers; automotive, wireless, and ITS OEMs; State and local DOTs; and U.S. DOT CV program managers who are overseeing safety applications work.

1.3 System Overview

The RCVW system leverages all the technologies and components developed by the current CV applications and includes additional capabilities to enhance motor vehicle safety when approaching or traversing an HRI. These capabilities include providing:

- in-vehicle information regarding activation status of the HRI,
- an in-vehicle warning when the RCVW system predicts imminent violation of an active HRI,
- integrated and improved communication technologies,
- improved vehicle locational accuracy,
- a cloud-based subsystem (CBS) for disseminating HRI activation status and HRI-related attributes, and
- road-based subsystem (RBS) operational status via the internet to authorized subscribers.

The main function of the system is to provide a warning to the vehicle driver of a predicted imminent violation of an HRI equipped with active warning/protective system device(s). Alerts and warnings that are both timely and effective in alerting vehicle operators, who otherwise may

be unaware of potential danger in their surroundings, are critical in the prevention of avoidable incidents.

A second function is to relay the HRI activation status and RBS operational status via the CBS to subscribed users such as vehicle navigation systems, commercial dispatchers, first responders, and Traffic Management Centers (TMC).

The RCVW system consists of three main and physically separate subsystems: a vehicle-based subsystem (VBS), an RBS, and a CBS. The RBS and VBS were designed, to the extent practical, employing common hardware and software components. Subsystem-specific elements were incorporated as needed. Data is exchanged between all three subsystems either by wired link, radio, or the internet. The system, in its entirety, includes the following subsystems and components:

- Computing Platform (CP) – The heart of the RCVW system is the CP, utilized in both the VBS and the RBS. The CP interfaces to the other supporting subsystems and includes a central processor and RCVW-specific software applications.
- Data Communication Radios – Radios will be utilized in both the VBS and the RBS. Radios and associated protocols will provide low-latency wireless communications between the two systems.
- CBS – The CBS will serve as an intermediary between the RBS and third-party application services to provide primarily: HRI activation status and RBS operational status via the internet.
- Data Communication Backhaul – The RBS requires connectivity to the CBS through a wide-area network (WAN) backhaul communication network (BCN). This would ideally be a low-latency fiber connection but is likely to include one or more wireless links. This connection is not provided by the RCVW system.
- Global Navigation Satellite System (GNSS) – A multi-band GNSS module will be utilized within the VBS to provide sub-meter positional accuracy. The positional augmentation data required to attain this degree of accuracy will be supplied from the RBS to the VBS via the Data Communication Radios. This augmentation data will be either produced within the RBS by means of a GNSS module or received from the CBS, depending on how the system is configured.
- Driver-Vehicle Interface (DVI) - The DVI will be utilized to provide the vehicle driver RCVW warnings and alerts.
- Software Applications – Each of the RCVW subsystems will implement specific software applications based on their specific functionality. The RCVW CV safety application will be enhanced to interface with the CBS to support a variety of new applications, particularly for traffic management and navigational use cases.

1.4 Interested Participants

Interested participants are the designers or manufacturers of ITS equipment and systems; installers of ITS equipment/systems; railroads; mobile devices application developers; and the agencies of Federal, state, county, and city/town/municipalities responsible for public safety.

1.5 Organization of the Report

The remainder of this document consists of the following sections and content:

- Chapter 2 lists the documents used as background information or as a source of user needs. Many of these documents are artifacts from previous ITS development programs.
- Chapter 3 provides an overview of the current system. This is used as the basis for analyzing the needs and capabilities to be considered in the revised system.
- Chapter 4 discusses the RCVW ConOps needs and the process followed to identify and define them.
- Chapter 5 describes the general concepts of the proposed RCVW System. The chapter includes system scope, operational environment, operational policies and constraints, major system services, and interfaces to external systems and subsystems.
- Chapter 6 discusses the impacts resulting from the proposed system, both beneficial and adverse.
- The end of the document contains an alphabetical listing of acronyms used throughout and their meanings.

2. Referenced Documents

2.1 Code of Federal Regulations (CFR)

- 49 CFR 234, “Grade Crossing Signal System Safety and State Action Plans”
- 49 CFR 236, “Rules, Standards, and Instructions Governing the Installation, Inspection, Maintenance, and Repair of Signal and Train Control Systems, Devices, and Appliances”
- 49 CFR 236, Docket No. FRA–2011–0028, Notice No. 3], RIN 2130–AC27 “Positive Train Control Systems”

2.2 U.S. Department of Transportation

- FHWA-JPO-16-408 “Vehicle-to-Infrastructure Rail Crossing Violation Warning: Concept of Operations”
- FHWA-JPO-18-609 “Vehicle-to-Infrastructure Prototype Rail Crossing Violation Warning Application. Project Report.”
- DOT FRA ORD-22/07 “Rail Crossing Violation Warning Application – Phase II”

2.3 Institute of Electrical and Electronic Engineers (IEEE)

- IEEE 1570 (2002) IEEE Standard for the Interface Between the Rail Subsystem and the Highway Subsystem at a Highway Rail Intersection
- IEEE 1362-98 (R2007), Guide for Information Technology, System Definition, Concept of Operations (ConOps) Document
- IEEE 1483-2000 IEEE Standard for Verification of Vital Functions in Processor-Based Systems Used in Rail Transit Control
- IEEE 1609 Standards for Wireless Access in the Vehicular Environment (WAVE)
- IEEE 1609.0-2019 – IEEE Guide for Wireless Access in Vehicular Environments (WAVE) Architecture
- IEEE 1609.2-2016 – IEEE Standard for Wireless Access in Vehicular Environments (WAVE) – Security Services for Applications and Management Messages
- IEEE 1609.3-2020 – IEEE Standard for Wireless Access in Vehicular Environments (WAVE) – Networking Services
- IEEE 1609.4-2016 – IEEE Standard for Wireless Access in Vehicular Environments (WAVE) – Multi-Channel Operations
- IEEE 1609.12-2019 – IEEE Standard for Wireless Access in Vehicular Environments (WAVE) – Identifiers

2.4 Society of Automotive Engineers (SAE)

- SAE J2735_202211 V2X Communications Message Set Dictionary
- SAE J1757/1_202108 Standard Metrology for Vehicular Displays
- SAE J2402_201001 Road Vehicles – Symbols for Controls, Indicators, and Tell-Tales

- SAE J2540/2 IT IS (International Traveler Information Systems) Phrase Lists

2.5 National Electrical Manufacturers Association (NEMA)

- NEMA TS 2-2021 v.03.08 Standard for Traffic Controller Assemblies with NTCIP Requirements

3. Current System or Situation

3.1 Background

A CV safety application such as the one described in this ConOps to increase safety at HRIs has been demonstrated but not widely deployed. Existing methods of HRI protection consist of passive and active warning devices and protective countermeasures. Active warning devices include flashing lights, bells, wigwags, and highway traffic control signals. Protective countermeasures include gates employed in a variety of site-specific configurations. Supplementary warning devices, such as train-activated, fixed, and variable message signs, have been installed on a limited basis.

Numerous techniques and technologies have been implemented to reduce the frequency and severity of crashes at HRIs. However, analyses of US DOT accident databases indicate that further improvement of the current systems is warranted. The proposed enhancements in this ConOps will supplement the current systems with different and innovative technology to further improve safety and mobility at HRIs.

3.2 Description of Current System and Operational Modes

All HRI active warning systems have three basic modes of operation:

1. The flashing lights and bells are not active, and the gates (if present) are in the raised position, indicating a train is not detected by the train detection system.
2. The flashing lights and bells are active, and the gates (if present) are in the lowered position, indicating a train is detected by the train detection system.
3. A fail-safe mode in which the active warning and protective devices are actuated as a result of a system failure.

Variable Message Signs (VMS) and Second Train Approaching signs, when present, are part of the active warning/protective system. Traffic signal preemption is sometimes provided to nearby signalized intersections where queues from the intersection or crossings can potentially impact each other. In these cases, the normal mode of operation of the traffic signal is preempted to operate in a special mode when a train is approaching.

3.3 Policy Influence on Safety Systems

The primary goal of the HRI safety application is to provide motorists with advance warnings concerning potentially unsafe conditions at HRIs. However, local, regional, state, and national policies all influence the ways in which these systems are designed, installed, operated, and maintained. Examples of the types of policy questions that may influence how these systems operate in a given area and need to be addressed prior to deployment are as follows:

- Should all types of users (e.g., light vehicles, commercial vehicles, pedestrians) be treated equally, or should one particular user class be prioritized over the others?
- Within a region, should all HRI warning systems have a consistent design and operation? Should the same warnings be provided throughout a region or state, or can different types of systems operate to provide different warnings?

- What federal and state-level standards, if any, need to be followed in the use of static and dynamic HRI signage?
- What federal and state-level regulations, if any, for experimental sign and warning systems are not addressed by current design standards?
- What sort of discussions and agreements need to take place with the Class I railroads, short line railroads, and commuter railroad regarding the addition of equipment on their property?

3.4 Hardware and Software Considerations

The use of cloud-based services and CV technologies to increase the safety of a motorist approaching an HRI requires addressing the following considerations:

- Power consumption of hardware components and internet connectivity when an off-grid deployment is required.
- Types, nature, and timing of driver alerts and warnings provided to motorists approaching the HRI.
- Ability of RCVW RBS to interface with 1) the preemption output(s) of all existing HRI equipment that is used to control HRI warning devices and 2) IEEE 1570-compliant serial communication devices.
- Positional accuracy of in-vehicle GNSS modules that do not receive Radio Technical Commission for Maritime Services (RTCM) corrections via a cellular module.
- A secure means and message protocol will be needed to push HRI activation status and RBS operational status to the internet.
- Current state of CV technology, with respect to its maturity and dependability
- Organizations required to apply for CV broadcasting licenses for planned system deployments.
- System defenses against cyber events and unauthorized communications access.
- Failure mode or degraded state operation of the cloud-based system segments.

4. Justification for and Nature of Changes

This ConOps focuses on defining user needs and the operating concepts for a safety application that utilizes track-circuit based or other train detection means to provide roadway-vehicle drivers with timely alerts and warnings.

4.1 Justification for Changes

The US. DOT FRA at-grade crossing safety statistics indicate that in 2022, a total of 201,056 HRIs were in service, of which 125,124 were public and 75,932 were private. The public HRIs were roughly divided evenly between active warning devices and passive warning devices. A total of 1,192 HRI incidents involving motor vehicles and trains occurred at HRIs. These incidents involved 152 fatalities and 419 injuries, resulting in substantial economic costs.

Additional protections for vehicle drivers traversing HRIs beyond traditional active warning devices are needed. Table 1 presents a review of FRA safety statistics since 2015³ that does not reflect appreciable improvement in HRI incidents involving motor vehicles (i.e., excluding pedestrian and other categories), fatalities, or injuries.

Table 1. HRI Safety Statistics for Calendar Year 2022.

Calendar Year	Motor Vehicle Incident	Fatalities	Injuries
2015	1,926	188	991
2016	1,873	175	811
2017	1,952	193	799
2018	2,029	177	780
2019	2,041	193	780
2020	1,728	124	611
2021	2,145	234	669
2022	1,192	152	419

Source: Federal Railroad Administration

While active warning devices such as flashing lights, bells, and gates are effective at decreasing risk, these HRIs still account for roughly one-half of all crashes. The statistics also show that the

³ Source: Federal Railroad Administration Office of Safety Analysis [website](#)

top causes attributed to HRI crashes are distracted drivers and driver judgment errors.⁴ Distracted drivers may not notice that they are approaching an HRI, or they may fail to perceive activated warning devices or to recognize that a train is approaching. Situational awareness of a vehicle operator may be challenged due to adverse atmospheric conditions, engagement with a myriad of distractions (e.g., cell phone use or personal interactions), the influence of alcohol/drugs/medication, or impaired mental capacity due to fatigue or medical events. Existing warning devices do not communicate with roadway-vehicle systems.

4.2 Description of Desired Changes

Modern instrumented vehicles and roadside electronics allow for the implementation of innovative applications to enhance motorists' situational awareness and reduce HRI incidents. These applications offer the potential to enhance safety and provide benefits in mobility and convenience to the traveling public. The desired change, therefore, is to increase safety, mobility, and awareness by integrating existing and emerging technologies within the context of a CV system and by providing a means to communicate with cloud-based services.

The RCVW system will provide warnings and alerts to vehicles approaching an HRI equipped with a train detection means. It combines real-time vehicle information, such as speed, type of vehicle, and location information, with HRI activation status to provide alerts and warnings if a violation is predicted. The CBS enhanced system will also enable the dissemination of timely HRI activation status to subscribers of cloud-based services. This information can be used by authorized subscribers to enhance existing traffic management and navigation applications. For example, an active grade crossing may result in nearby motorists making changes in their route, enabled by their navigation system or TMCs relaying real-time information to the public on the activation status of an HRI.

4.3 Factors Considered but not Included

Storage Distance⁵ is not a requirement for this ConOps and is addressed by an existing traffic control signal preemption algorithm. The Manual of Uniform Traffic Control Devices (MUTCD) states in Section 8C.09: "If a highway-rail grade crossing is equipped with a flashing-light signal system and is located within 200 feet of an intersection or midblock location controlled by a traffic control signal, the traffic control signal should be provided with preemption."

4.4 Assumptions and Constraints

- The proposed system is only fully compatible with HRIs with active warning/protective devices such as gates, bells, and flashers (i.e., having a train detection system).
- Existing active HRI warning protective devices do not communicate with roadway-vehicle systems.
- Existing active HRI protective devices do not broadcast activation status over a public network.

⁴ Source: FHWA-JPO-18-609 "Vehicle-to-Infrastructure Prototype Rail Crossing Violation Warning Application. Project Report.

⁵ Storage Distance is defined as the distance available for vehicle storage measured between 6 feet from the rail nearest a traffic intersection to that intersection's stop line or the normal stopping point on the highway.

- The interface between the proposed RCVW system and the track-circuit based train detection means the system will be the direct current (DC) voltage (with return). The interface between the HRI controller/IEEE 1570-compliant device and the RCVW system will be a serial communication link that provides preemption status.
- The integration of the CBS with the in-vehicle safety application proposed in this ConOps has the potential to provide better situational awareness to roadway-vehicle operators.
- The full deployment of CV technology in roadway vehicles, especially non-commercial ones, is expected to take many years.
- The majority of vehicle users have access to a mobile device connected to the internet, which provides a potential recipient of the cloud data.
- Traffic management and navigation applications currently exist but require timely and dependable data inputs such as HRI activation status to be more effective.

4.5 Stakeholders

- Railroads – Class I railroads, short line railroads, and commuter railroads
- Motorists – private, commercial, federal, state, and municipal
- Traffic control equipment manufacturers
- Vehicle original equipment manufacturers
- Railroad signal equipment suppliers
- State DOT agencies
- Mobile application developers
- Commercial trucking dispatchers

A list of stakeholder needs is found in Table 2. The table provides a list of needs and their scope. The scope is categorized as:

- General – Need that involves the system as a whole (including user)
- VBS – Need that involves the on-board subsystem
- RBS – Need that involves the road-based subsystem
- CBS – Need that involves the cloud-based subsystem

The table of stakeholder needs has been updated from previous revisions of this ConOps; some needs have been removed to better reflect the current operation of the system. Numbering remains for traceability.

Table 2. Stakeholder Needs

No.	Title of Need	Scope	Description of Need
N-001	Driver Population	General	Flexibility to consider and adjust for the full range of drivers and their capabilities, including but not limited to inexperienced (e.g., recently licensed) drivers, distracted drivers, and older adult drivers (e.g., slower reflexes,

No.	Title of Need	Scope	Description of Need
			impaired hearing, loss of peripheral vision, diminished eyesight, or otherwise physically impaired)
N-002	Human Factors	General	Warnings are effective and compatible with automotive human factors guidelines, OEM driver-vehicle interface principles and practices, and driver-vehicle interfaces that follow human factor guidelines by the Federal Highway Administration (FHWA), National Highway Safety Administration (NHTSA), and SAE
N-003	Vehicle Compatibility	VBS	Suitable for all vehicle classes and types
N-004	Vehicle Data	VBS	Speed and vehicle performance data related to braking
N-006	Operating Environment	VBS RBS CBS	Operates in all weather, road surface, and visibility conditions
N-007	Positional Accuracy	VBS	Accurate position data
N-008	HRI Data	RBS VBS CBS	Correct and timely, HRI activation status and configuration data
N-009	Deployment Sites	VBS CBS	Effective operation at all HRIs, regardless of HRI configuration, number of tracks, or skew
N-010	Relative Position Determination	VBS	Provides lane-specific warnings
N-011	Performance Area	VBS RBS CBS	Effective operation in urban, suburban, sub-rural, and rural areas
N-012a	Infrastructure Compatibility	VBS RBS	Interoperates with current infrastructure safety systems (e.g. traffic control and train detection systems) in accordance with NEMA TS 2-2021 v03.08
N-012b	Serial Interface Compatibility	VBS RBS	Interoperates with the preemption output(s) of all existing HRI equipment and IEEE 1570-compliant serial communication devices
N-013a	V2X Communications	RBS VBS	Communicates via Dedicated Short-Range Communications (DSRC)and/or C-V2X in compliance with the IEEE1609 suite of protocols
N-013b	Backhaul Communications	RBS CBS	Communicates via Transmission Control Protocol/Internet Protocol (TCP/IP) to the CBS over an established BCN
N-013c	Communications	CBS	All messages from RBS(s) are time stamped
N-013d	Communications	RBS	Continuously verify RBS-CBS link communications

No.	Title of Need	Scope	Description of Need
		CBS	
N-014a	Communications Reliability	RBS VBS	Ensure that RBS transmitter and VBS receiver failures are detectable
N-014b	Communications Reliability	RBS VBS CBS	Ensure that the RBS is reporting HRI geometry and HRI activation status at the proper rate
N-14c	Positional Reliability	VBS	Ensure that VBS positional accuracy is within required margins
N-015	HRI Safety	RBS	Upon removal of a discrete DC voltage preemption signal (s) or receiving an IEEE 1570-compliant serial communication message from an HRI controller, broadcasts HRI active message to the VBS and the CBS. Broadcasts HRI inactive message when the HRI controller restores the preemption signal, or when the IEEE 1570-compliant serial communication device sends the message that the preemption signal has been restored.
N-016a	Message Integrity	VBS	Issues onboard safety warnings when warranted
N-016b	Message Availability	RBS VBS CBS	Issues HRI status messages, crossing geometry and ID periodically
N-017	Onboard Warning	VBS	Warns drivers of potential violation-imminent situations with an oncoming train or a train occupying the HRI, in time for the driver to take appropriate action
N-018a	Onboard Alert	VBS	Alerts drivers of an active HRI ahead
N-018b	Cloud Alert	CBS	Alerts cloud subscribers of HRI protective devices activation/deactivation
N-019a	Fail Safe Design	RBS	Equipment and system design to incorporate provisions to prevent false safe indications
N-019b	False Negative Assurance	VBS RBS	Nominal operation with no possibility of missed warnings
N-020	Self-Diagnostics	VBS	System informs the driver via DVI when critical components are not reporting data
N-021	Status Reporting	RBS CBS	Reports the operational status of any of its infrastructure components to the CBS when a component is offline, such as the result of a self-diagnosed fault
N-023	Site Security	RBS	Prevents unauthorized access to the RBS shelter
N-024	Notification of Intrusion	RBS CBS	Reports to a central maintenance facility when unauthorized access to the RBS shelter has occurred, or attempted

No.	Title of Need	Scope	Description of Need
N-025	Communications Security	RBS VBS CBS	Prevents unauthorized access to the system intelligence (e.g., hacking, spoofing), and verifies the authenticity of all messages between the RBS – VBS and RBS – CBS
N-026	Connected Vehicle Security Performance	RBS VBS	Implementation of secure-communication protocols perform within the boundaries of a vehicle safety application
N-027	Interoperability with other Onboard Systems	VBS	Does not interfere with automotive industry safety systems
N-028	System upgrades	RBS VBS CBS	Future upgrades in VBS software do not adversely affect existing RBS applications. Future upgrades in RBS do not adversely affect VBS/CBS applications. Future upgrades in CBS software do not adversely affect existing RBS applications
N-029	Maintenance	RBS	Roadside infrastructure (including a link to a BCN) will need to be maintained
N-030	Reliability Maintainability Availability	RBS VBS CBS	Employs commercially available system components that comply with the system design objectives and provide optimum cost-effective system availability
N-031	Environmental	RBS	Employs system components that comply with the railroad industry's environmental requirements
N-033	Configuration Management	RBS	All software and equipment modifications to the RBS must be approved at the design engineer level and <u>all</u> affected documentation must be revised and distributed accordingly
N-034	RBS/CBS Link-Integrity Monitoring	RBS CBS	RBS/CBS link will be monitored to ensure that a "sufficient" number of SPaT and MAP messages are being received

4.6 Operational Need

As previously mentioned, traditional warning devices are limited in effectiveness when a motorist's situational awareness is compromised. It is, therefore, reasonable to assert that there is an operational need to enhance drivers' situational awareness when approaching an HRI if improvements to safety are to be realized.

Situational awareness of a roadway-vehicle driver may be less than ideal due to:

- Adverse atmospheric conditions when sight and/or hearing are limited – dense fog, intense precipitation, electrical storms, sand/dust storms, solar glare, etc.
- Distractions from any number of sources, e.g., texting, cell phone use, roadway hazard avoidance, emergency maneuvers, vehicle malfunction, personal interactions with a child or other adult, etc.
- The influence of alcohol, illegal substances, or medication.
- Impaired mental capacity due to mental or physical fatigue or medical condition.

The intent of this ConOps is to define a concept that provides a motorist approaching an HRI in-vehicle multi-sensory warnings/alarms designed to overcome the effects of compromised or impaired situational awareness. These are to be generated as a function of vehicle location, direction of travel, speed of the vehicle, and activation status of the HRI.

Improved situational awareness can also be beneficial in the vicinity of an HRI that is not directly HRI safety related. For example, any incoming traffic that can effectively be re-routed beyond the active crossing would help alleviate queues and help establish unimpeded emergency vehicle response paths. Likewise, real-time monitoring of the active HRI at a TMC may not only improve the incident response, should one occur, but could also substantiate the need for new or additional data points for the historical record of such incidents. A secondary goal of this ConOps is to provide a common framework for enabling other HRI-related applications.

5. Concepts for the Proposed System

The proposed system allows for the timely broadcast of the activation status of the warning devices of an HRI. This information is broadcasted simultaneously over radio for incoming CV's usage and distributed to subscribed users via the cloud. Approaching CVs will be able to use the information to warn the motorist on the activation of the crossing or of a predicted imminent violation of an HRI. Subscribed users such as TMCs, commercial dispatchers, first responders, and vehicle navigation companies will be able to receive timely information on the activation of an HRI over the internet; this information can then be used to plan alternative vehicle routes and generate alerts. A timely and effective warning to vehicle drivers and the public at large is critical in preventing incidents.

5.1 Background, Objectives, and Scope

Numerous techniques and technologies have been implemented to reduce the frequency and severity of crashes at HRIs. However, analyses of US DOT accident databases indicate that the current solutions do not sufficiently mitigate the risk found in the HRI environment.

This document presents an approach for implementing an HRI safety application that is train detection technology agnostic. It leverages existing track-circuit based, or other, train detection technology that, when integrated with the CV roadside architecture and cloud computing, constitutes a V2X approach to improving HRI safety.

Given that HRI characteristics vary widely, it is not the intent of this ConOps to discount the possibility of other approaches employing CV technology.

The primary objective is to deploy a CV system to attain the critical goal of providing a timely in-vehicle alert when there is a predicted imminent violation of an active HRI. In addition, implementing cloud computing technology will provide the activation status of the HRI to other authorized clients of the cloud. This technology allows the dissemination of HRI activation status and RBS operational status to remote locations and services. The success of this concept is highly dependent on vehicle-operator and public acceptance, which in turn is dependent on the reliable and predictable performance of all system elements, including:

- Track-circuit based, or other, train detection system and its associated IEEE-1570 serial communication device, where present
- RBS
- RBS to VBS Communications⁶
- RBS to CBS Communications⁶

Situational awareness on approach to an HRI can be increased or enhanced by providing roadway vehicles with timely multi-sensory VBS warnings/alerts of:

- An alert of an HRI with activated warning devices ahead
- A warning of a predicted and imminent violation of an HRI

⁶ The IEEE 1609.3 standard defines authentication and encryption criteria for V2X messaging to minimize the probability that the integrity of a Connected Vehicle's communication system is compromised.

Real-time HRI activation status cloud data can be used by local agencies, commercial entities, first responders, and navigational companies for safety and mobility applications.

The system is intended to be deployed at any HRI where benefit would be accrued by increasing situational awareness to minimize safety-related incidents.

5.2 Operational Policies and Constraints

Information regarding train location and speed is outside of the scope of this ConOps. Railroads own and operate private wireless data networks that transmit safety-related and potentially sensitive operational data. As such, the operational policies of railroads regarding data sharing may limit the ability of public sector agencies to access data. Since each railroad is likely to have a different policy governing data access, public sector agencies may be required to negotiate data sharing agreements with each operating property. Public sector agencies involved in the deployment of RBSs on railroad property will likely require railroad authorization prior to installation.

5.3 Description of Operational Scenario and System

The operational environment described herein addresses HRIs for which the warning devices such as bells, gates, and flashers are activated by a track-circuit based, or other, train detection system. Implementing this scenario requires that the existing track-circuit based, or other, train detection system provide a discrete DC voltage preemption signal(s) or an IEEE 1570-compliant serial communication protocol message that indicates the approach, or the presence at an HRI, of a rail vehicle. HRIs are equipped with an RBS which broadcasts the activation status of the HRI (active or inactive), the location of the crossing, its geometry, and the approach zone.

The two possibilities for this scenario are:

- An RBS is installed in the HRI.
- The existing track-circuit based, or other train detection system provides a discrete DC voltage preemption signal(s), or IEEE 1570 serial communication protocol messaging; in this scenario, add an application with HRI specific logic/messages to the existing RBS.
- The existing track-circuit based, or other train detection system does not provide a discrete DC voltage preemption signal(s) or IEEE 1570 serial communication protocol messaging; in this scenario, install the signal preemption control line(s) and add an application with HRI-specific logic/messages to the existing RBS.
- There is no existing RBS; an HRI RBS is to be installed with the application cited immediately above.

A flowchart of this scenario can be seen in Figure 1.

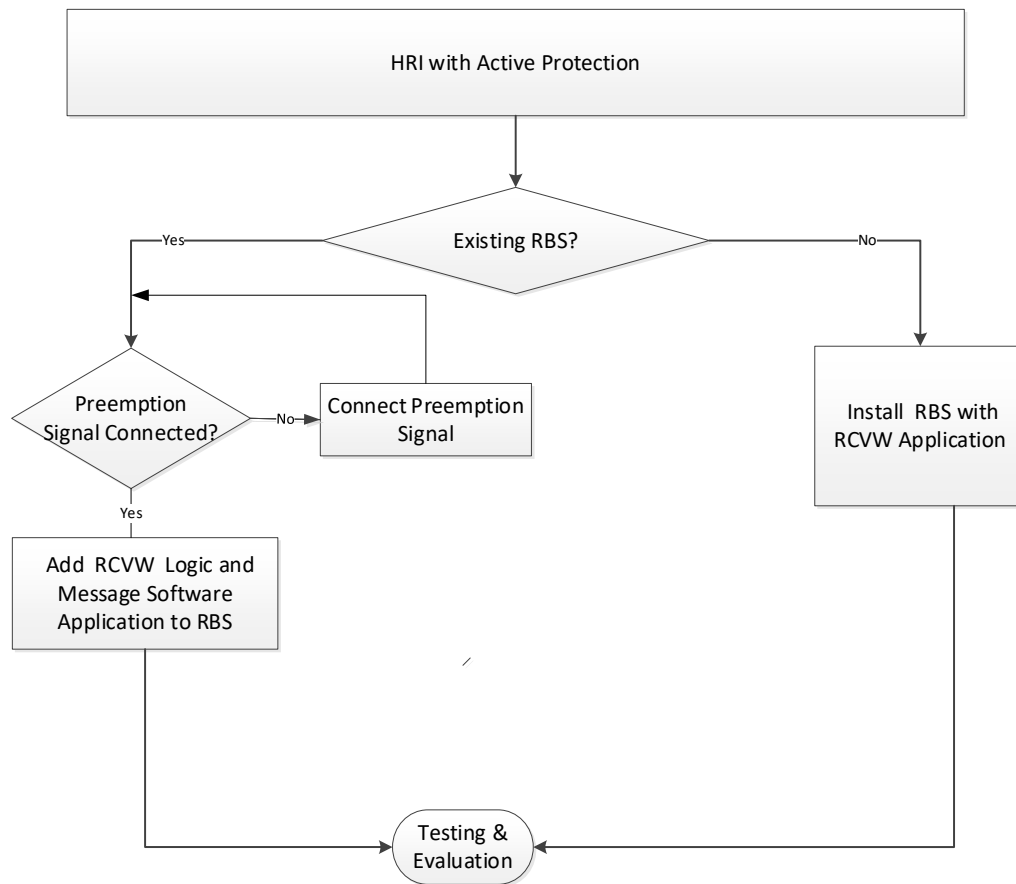


Figure 1. Flow Chart for Implementing RCVW Protection at an HRI with a Track-circuit Based Train Detection System

Source: Battelle

An HRI equipped with an RCVW-equipped RBS will broadcast the activation status of the HRI, HRI identification number, and GPS coordinates over radio⁷ to the VBSs within mission-effective range and to the CBS via the BCN. When an approaching train activates the HRI train detection system, the warning devices will be activated, in accordance with federal law, a minimum of twenty seconds prior to the arrival of the train. The RBS will then change the activation status of the HRI in the dataset which it is constantly broadcasting. This message is received by the roadway-vehicle VBS when it is within its mission-effective range of the RBS transmitter and the CBS. A block diagram of this operational scenario is shown in Figure 2. The direction of data flow for the message is one-way from the RBS to the roadway vehicle and to the CBS.

⁷ This data has been defined as Signal Phase and Timing (SPaT) and MAP in standard SAE J2735. SPaT is broadcasted every 100 milliseconds and MAP every 1 second.

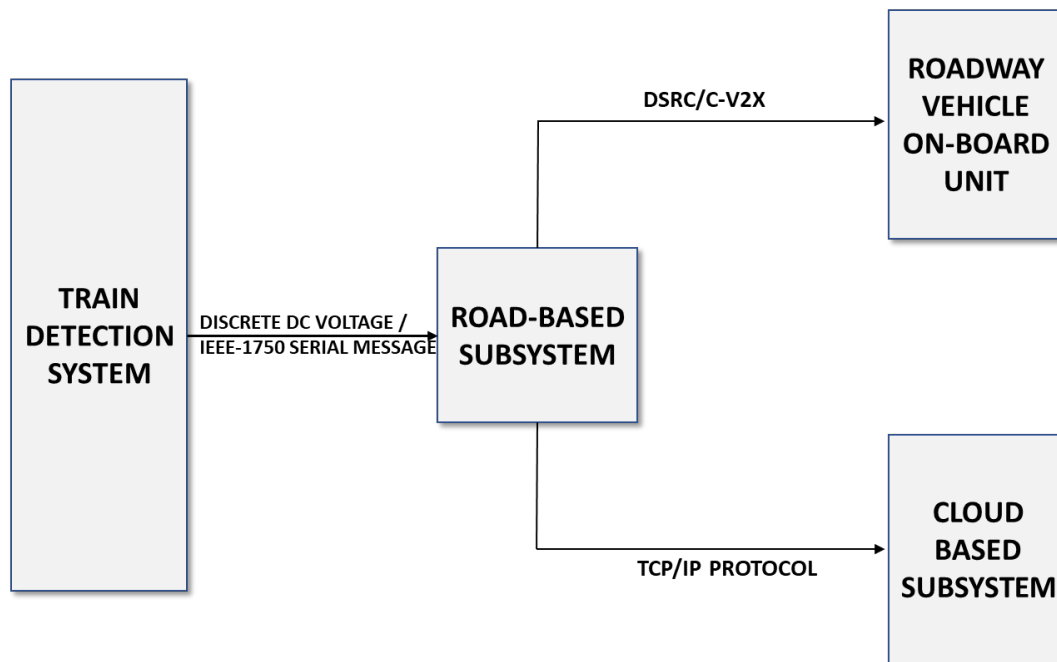


Figure 2. RCVW Block Diagram

Source: Battelle

5.4 Description of the Proposed System

The goal of the system is to increase the awareness of a driver approaching an HRI taking into consideration the vehicle speed and position, driver actions (i.e., deceleration rate and reaction times), and efficacy of in-vehicle alerts. The proposed system uses two distinct technologies to achieve the goal: one is the use of a low-latency CV technology to support the local safety messaging, and the second is by relaying information via the cloud. The CV system is described here:

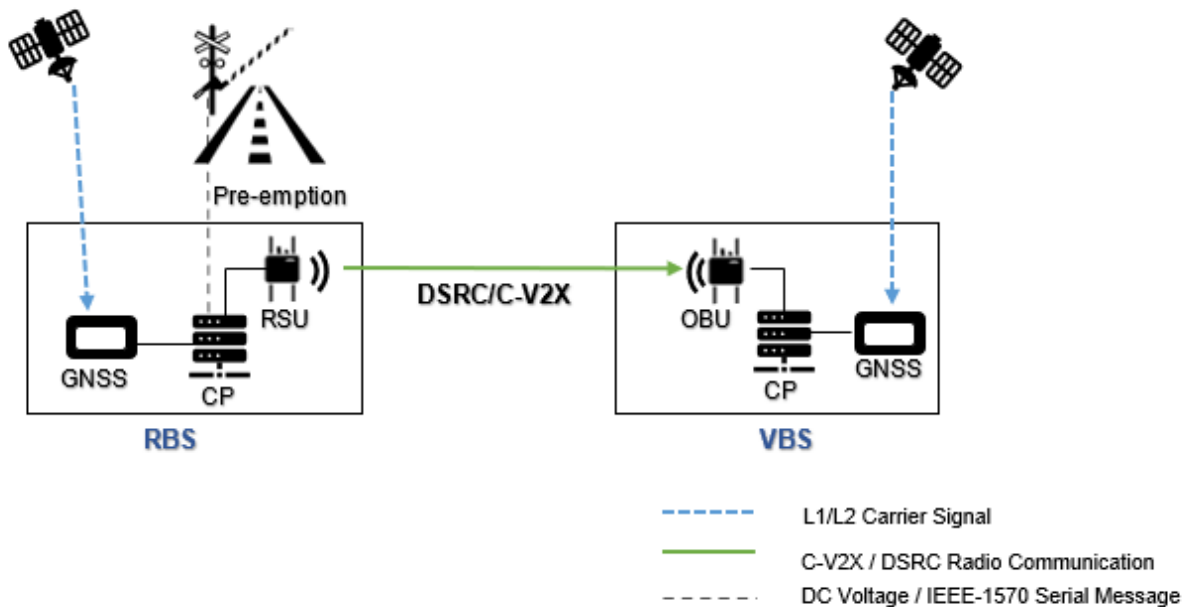


Figure 3: RCVW CV Local Safety Messaging

Source: Battelle

1. An HRI equipped with the RCVW system constantly broadcasts over radio the following HRI-specific data⁸:
 - a. Map Data (MAP) which includes approach zone and HRI geometry.
 - b. Signal Phase and Timing (SPaT) data which includes HRI activation status (i.e., whether an HRI is active or not).
 - c. RTCM corrections (to be used by a vehicle onboard GNSS). These corrections are generated within the RBS by means of a GNSS module serving as a local RTCM base station.
2. A vehicle equipped with the RCVW system enters the mission-effective range of a known RBS, and its VBS begins processing the HRI-specific data it receives. At this point, system fault processing initiates.
3. The vehicle enters the approach lane MAP.
4. If the vehicle VBS receives and processes data from the RBS indicating that the HRI is not active, no alerts or warnings are issued, and the system continues to process the received data.
5. If the vehicle continues within the approach lane, and its VBS receives and processes data indicating that the HRI is active, the system will:
 - a. Issue an audio and graphic alert to inform the driver of an active HRI ahead.

⁸ The system follows the SAE J2735 and IEEE 1609 Standards.

- b. Continually determine the vehicle position relative to the HRI.
- c. Correlate vehicle speed and performance parameters and position with respect to the HRI to assess the probability of a violation.
- d. Issue an audio and graphic warning if a violation is imminent (i.e., system algorithm detects that the vehicle's distance to the HRI is not sufficient to allow the vehicle to stop before reaching the stop bar, when applying nominal braking, at its current speed).

Continue to correlate vehicle speed and performance parameters versus position with respect to the HRI to determine whether the driver is responding to the in-vehicle warnings. If the driver performs corrective actions (i.e., braking), it will extinguish the warning message and revert to an inform alert (i.e., if the HRI is still active) or extinguishes all alerts when HRI is no longer active. In addition, the system issues a warning when the vehicle is detected to be within the HRI, alerting the driver to clear the tracks.

The described CV approach requires RBS access to the HRI to acquire the current HRI activation status and an RBS device having a broadcast radio to transmit activation status directly to a VBS. However, this strategy has some limitations, including:

1. Limited deployment of this specialized equipment, particularly within vehicles
2. Licensing and deployment challenges for 5.9GHz band radio communications

Incorporating a CBS in the RCVW system provides the means to relay the HRI activation status through the internet via a BCN. The intent of this is not to replace the traditional CV architecture and functionality but to develop alternative communication mechanisms in order to potentially reach drivers of vehicles that are not ITS CV equipped, as well as authorized subscribers. Data can be shared between disjointed devices through an intermediary in the *cloud*, which is generally a Software-as-a-Service (SaaS) offering that facilitates connectivity through publicly accessible virtual infrastructure. For the proposed system, the CBS would operate on existing cloud-service providers. The CBS will receive the same messages as a VBS. The CBS, upon detecting an HRI status change (active/inactive), pushes it down to authorized partner application services. These partners can include navigation and commercial freight dispatch systems, and the information can be used to help plan routes around active HRIs, as well as to alert drivers to an approaching train. In addition, the CBS allows the broadcast of correction data from the Networked Transport of RTCM via Internet Protocol (NTRIP) casters to the RBS. This provides the RBS the flexibility to choose whether to produce the RTCM corrections or receive them via the internet for subsequent broadcast over the radio.

The inclusion of a CBS expands the potential capabilities of the system beyond the HRI activation status application. The CBS could be used to compile and broadcast attributes of any HRI in the network. An authorized user would be able to retrieve specific information such as HRI location, its activation status, low-ground-clearance notification, incident statistics, and any other available information.⁹

⁹ This additional functionality is outside of the scope of this ConOps and is provided for reference only.

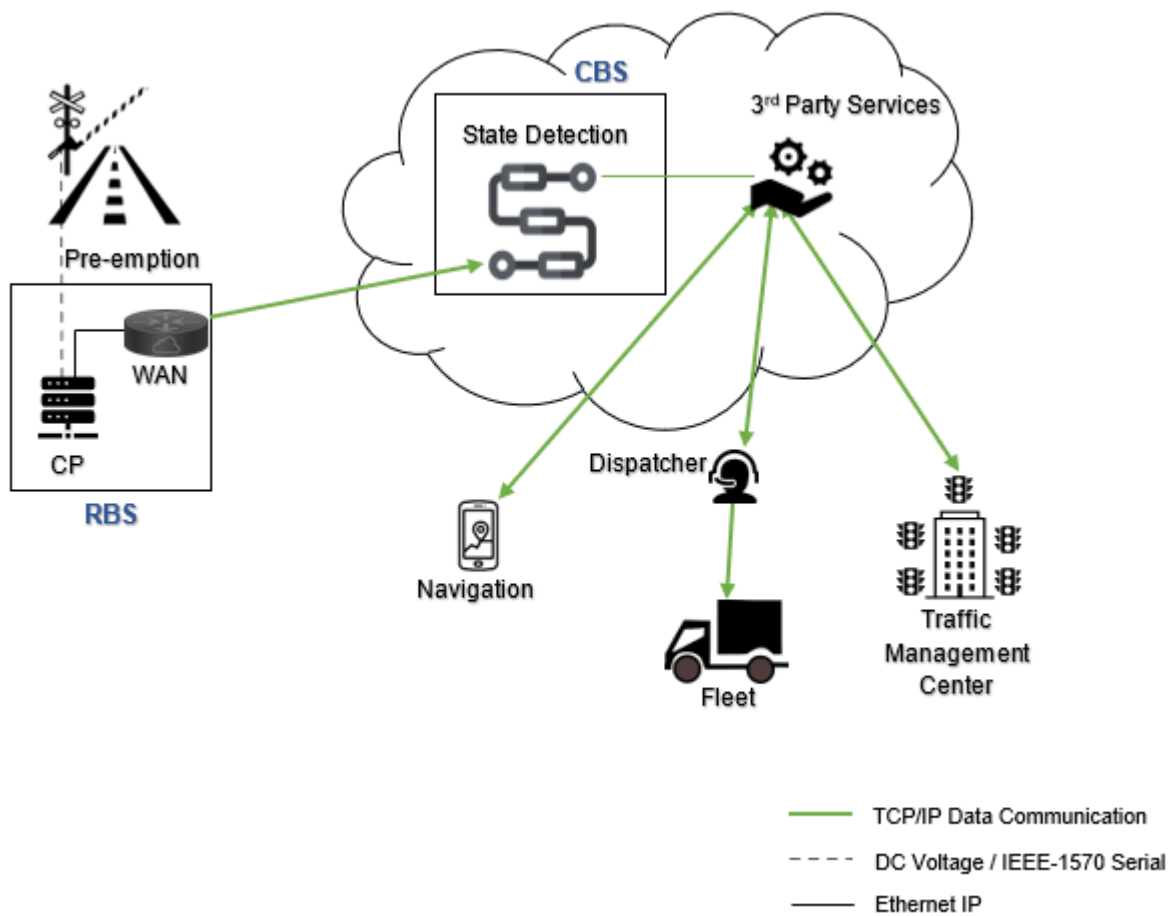


Figure 4: RCVW CBS HRI Activation Status Messaging

Source: Battelle

Figure 4 depicts the HRI activation status and RBS operational status communications from the RBS to the CBS. Note that there are two separate ends to the cloud-enabled communications. First, the WAN backhaul from RBS to CBS should be a reliable and secured link designed for a high volume of *internet of things* (IoT) traffic. Secondly, the downlink out of the cloud services to the individual applications will rely primarily on the wireless service providers; thus, the availability and reliability of the communications is not directly in the CBS scope. The CBS HRI activation status messaging can be summarized by the following:

1. The SPaT¹⁰ (along with associated MAP) message is pushed from the RBS to the CBS over the BCN.
2. The CBS utilizes existing ingest functions in the cloud provider to receive messages from RBS.

¹⁰ Note that each raw telemetry message is to be distributed to the cloud, not only the HRI activation status changes. This is done purposefully to ensure a complete audit log of events and so the CBS does not infringe on any existing proprietary systems that track only change-of-state events.

3. Received messages are stored in a cloud data store.
4. Change in HRI status from inactive to active (or vice versa) is detected by the CBS, and an HRI Activation Status Change message is created.
5. HRI Activation Status Change message is relayed to a third-party cloud service either as a result of a polling request or via an asynchronous broadcast to its web URL.
 - a. Any subscribed-user service receives this information.
 - b. A navigation system subscriber receives notice that the vehicle must stop for the train or be re-routed around the HRI.¹¹
 - c. An application running on a mobile device receives notice that the device is approaching an active HRI and issues an *inform* alert to the device.¹²
 - d. A subscribed commercial or emergency dispatcher receives notice and relays updated information to potentially affected fleet vehicles in the vicinity.
 - e. A subscribed traffic management center receives notice, and *it* initiates additional operational activities¹² to monitor for secondary traffic incidents.

The current CV safety messaging illustrated in Figure 3 requires that the VBS receive GNSS positional augmentation to meet the RCVW performance requirements. This is presently achieved through a local base station in the RBS that is set up to generate RTCM corrections that is then shared with the VBS. While this provides a highly available solution, a more convenient option may be to obtain the RTCM messages through an active connection to an NTRIP service such as the Continuously Operating Reference Stations (CORS). The addition of the CBS to RCVW provides a highly automated option for receiving RTCM corrections that the RBS can then share with the VBS, which does not require complex surveying techniques. It should be noted, however, that the RCVW RBS is expected to provide configuration options to deploy with either a base station or using the CBS as an NTRIP proxy. The RTCM proxy messaging is described below:

¹¹ Functionality is outside the scope of this ConOps.

¹² Functionality is outside the scope of this ConOps. Scenario is being researched for feasibility.

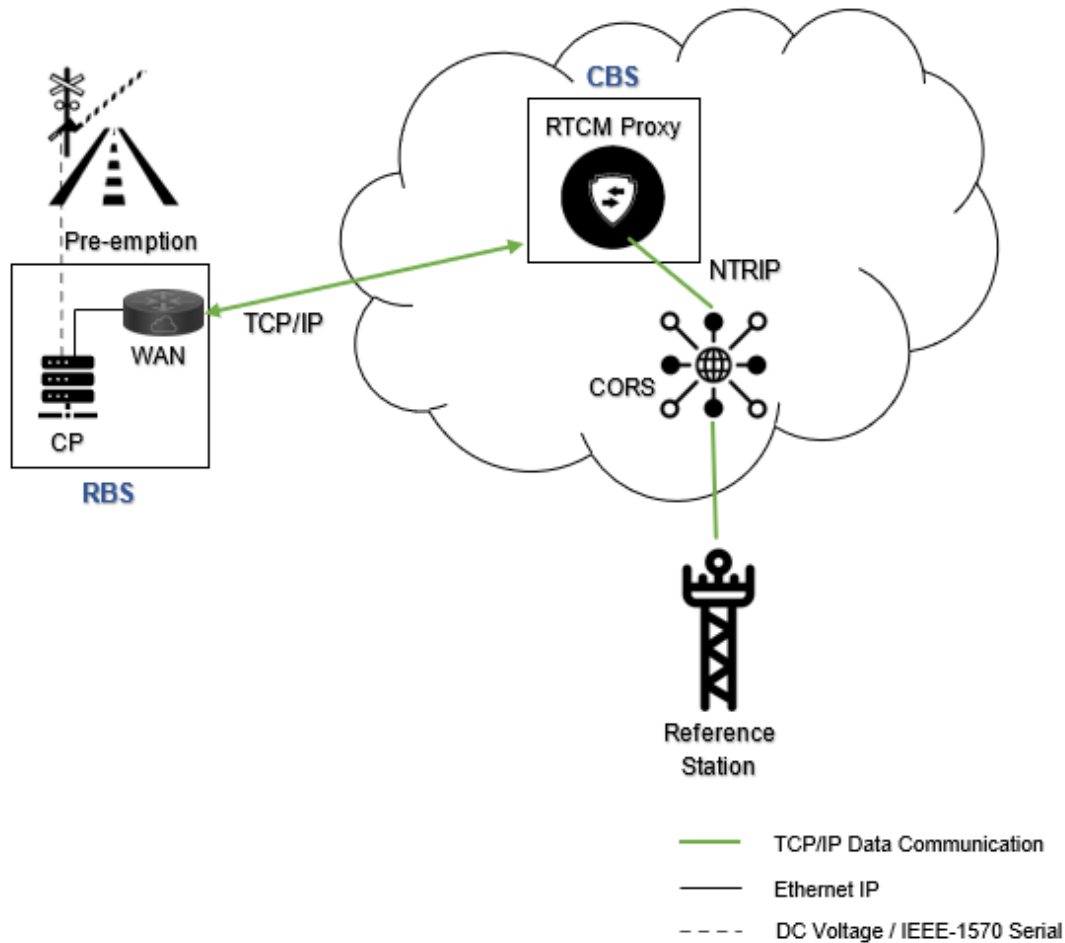


Figure 5. RCVW CBS GNSS Correction Messaging

Source: Battelle

1. The MAP message is pushed from RBS to CBS over the BCN.
2. The HRI's GPS coordinates are used by the CBS to acquire and maintain current corrections data from the Networked Transport of RTCM via Internet Protocol (NTRIP) caster.
3. The CBS receives RTCM messages.
4. The CBS relays the RTCM messages to the RBS over the BCN.

Finally, the CBS may be used as a secure data repository for HRI information. The CBS will provide Application Programming Interface (API)s for authorized subscribers to obtain static information about the HRI, such as location, road grade, low-ground-clearance, equipment, and software versions. The source of some of this information will be RBS MAP files, while more details can be added utilizing the US DOT Highway/Rail Crossing Database.¹³ The HRI information messaging is described below:

¹³ <https://safetydata.fra.dot.gov/MasterWebService/PublicApi/>

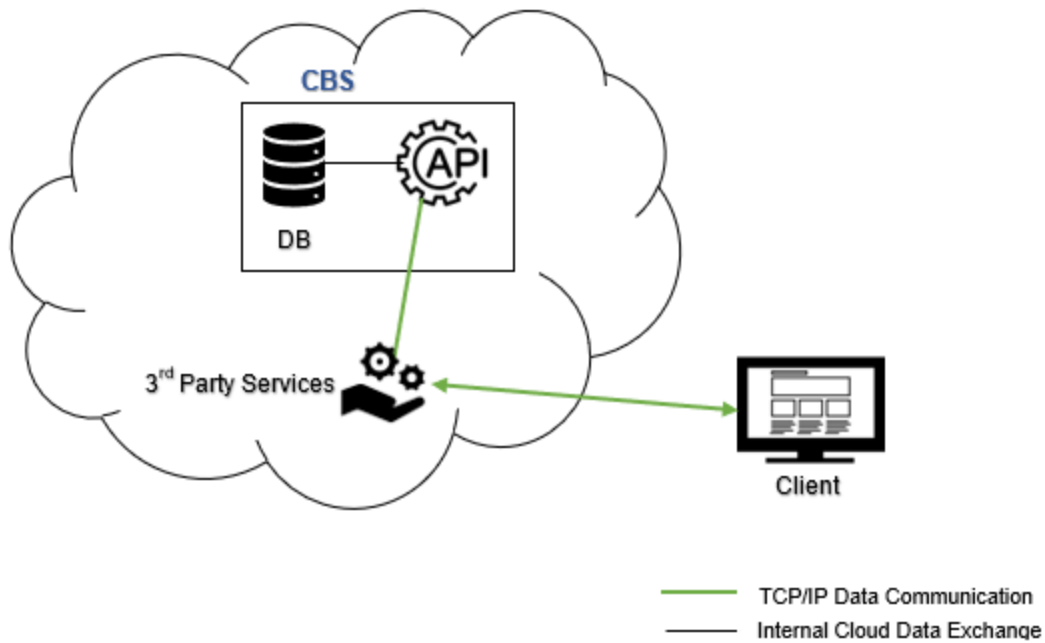


Figure 6: RCVW CBS HRI Information Messaging

Source: Battelle

1. A client, such as a web page or map, requests HRI data.
2. The cloud-based subscriber service forwards an API request to the RCVW CBS.
3. The RCVW CBS retrieves pertinent information from its database.
4. The resulting data reply is relayed back to the cloud-based subscriber.

For each scenario, communication latency is critical. The timely communication of HRI activation status information and the generation of alerts is essential for the success of the safety application.

Vehicles not having an RCVW VBS might access the RCVW CBS to acquire HRI-related data to gain significant advantage over the status quo. As an example, authorized navigation system service providers would acquire pertinent data, such as the status of the warning devices of an HRI en route. These applications would necessarily be developed by subscribers to the RCVW CBS database service; as such, they are not within the scope of the SRS associated with *this* ConOps. However, their potential benefit in the future further supports the establishment of an RCVW CBS.

5.5 Operational Mode

The normal mode of operation is defined as all subsystems and communication links functioning within specifications to enable a violation warning, or alert, to be generated by a roadway vehicle VBS and/or any subscribed user to the cloud service. The system is in normal operation if:

- The HRI activation status information is provided to the VBS and CBS in a timely manner.
- Connectivity of the RBS to the CBS is verifiable.
- The HRI geometry information is provided to the VBS and CBS in a timely manner.
- The VBS is able to obtain accurate and consistent GPS positioning in a timely manner.
- There are no opens or shorts in the cable between the RBS and HRI controller. Any deviation from what is defined as normal operation constitutes a failure mode and is reported to the user.

5.6 User Types and Other Involved Personnel

Users of the HRI CV safety application include the organizations, agencies, and individuals that are necessary for installing, maintaining, operating, and interacting with functioning CV applications. The primary users of the applications are:

- Roadway-vehicle OEMs – responsible for original equipment, vehicle-related equipment, and software actions necessary to establish and maintain the VBS.
- Aftermarket VBS manufacturers – responsible for hardware and software retrofits.
- State and local governments and their DOTs – responsible for installation and maintenance of CV roadside and wayside infrastructure.
- US DOT – responsible for providing guidelines to state and local agencies in the deployment and operation of CV safety applications.
- Motorists – responsible for the decisions made when approaching and entering an HRI.

Motorists are also responsible for the following:

- Familiarization with the vehicle safety features.
- Vehicle maintenance, especially of the VBS components.
- Assessment of the information provided by the VBS.
- Railroad signal equipment suppliers – responsible for the development and maintenance of railroad signal equipment that interfaces with CV applications.
- Traffic control equipment manufacturers – responsible for the development and maintenance of infrastructure equipment and software that can interface with CV applications (and other related safety systems, as they are fielded).
- Organizations responsible for CV safety application guidelines and standards – responsible for rules and procedures necessary for CV safety applications and components to become operational.
- Railroads (Class I, short line, and commuter railroads) – responsible for operation and maintenance of RBS.¹⁴

¹⁴ Possibly dependent on the physical location of RBS installation.

- Transportation Planners – Analyze HRI accident data to support decisions regarding the deployment of HRI CV infrastructure at additional locations.
- Cloud computing companies – responsible for software actions necessary to collect the information provided by the CBS and distribute it to authorized subscribers.

The need for information exchange between individual vehicles and roadside equipment may require the establishment of new working relationships among those organizations responsible for the design, development, operation, and maintenance of vehicles, software applications, and roadway systems.

Interaction between these organizations has, to date, been facilitated through groups such as the American Association of State Highway and Transportation Officials (AASHTO), American Railway Engineering and Maintenance-of-Way Association (AREMA), American Association of Railroads (AAR), Institute of Transportation Engineers (ITE), NEMA, SAE, and American Association of Motor Vehicle Administrators (AAMVA).

6. Impacts of the Proposed System

Implementing the HRI-related safety applications addressed by this ConOps will result in both beneficial and adverse impacts. The subsections below identify these principal resultant impacts.

6.1 Operational Impacts

The potential beneficial operational impacts, though few, are significant:

- Reduction of the number and severity of HRI safety-related incidents with an attendant reduction in societal costs.
- Potential improvement in emergency vehicle response time via in-vehicle dynamic routing when an HRI is to be occupied for an extended period of time.
- Improved traffic flow through the use of the cloud service.
- A system that is deployable nationwide will continually accrue benefits as an increasing number of roadway-vehicle drivers recognize its utility and adjust their driving behavior in response to its prompts.

The potential adverse operational impacts are also significant and will require much consideration when defining specifications:

- There is the potential risk that motorists may ultimately rely on the in-vehicle audio/visual display to alert them of approaching trains rather than the existing signs, signals, and pavement markings. It is, therefore, important to devise some means to indicate when VBS devices are not fully functional.
- There is also the risk of undetectable failures, e.g., for signals not sent on a periodic or continuous basis, it is not possible to detect a failure in the transmitter-receiver link if there isn't periodic testing. No signal being received could be due to the lack of a signal or a receiver failure.

6.2 Driver Impacts

The impact upon the driver in the deployment of V2X safety applications is enhanced situational awareness via warnings provided by the VBS. The specific means chosen to enhance in-vehicle situational awareness have been previously researched and published in several human factors design guidelines.⁽¹⁵⁾⁽¹⁶⁾ The application described in this ConOps will follow the visual and auditory message characteristics described in such guidelines.

6.3 Standards Organizations Impacts

Standards development organizations, such as IEEE, SAE, and NEMA, will be called upon to develop, maintain, and publish standards and guidelines in the following areas:

- Performance specifications, including reliability, maintainability, and availability
- Operations and maintenance

¹⁵ Campbell et al.(2016). Human Factors Design Guidance for Driver-Vehicle Interfaces

¹⁶ Campbell et al. (2004).In-Vehicle Display Icons and Other Information Elements, Volume I: Guidelines.

- Training and training materials for both operational and maintenance personnel
- Configuration management for both equipment and software/firmware
- Issuance of field change orders for implementing revisions (operating standards, equipment, or software/firmware) that impact system safety, performance, and reliability/maintainability
- Operational testing and evaluation to include certification
- Migration strategies for addressing obsolescent technology

6.4 Agency Impacts

The following are costs related to the implementation of this system that agencies may experience during implementation and deployment of the system:

- Site preparation
- Acquisition of RBS components
- Installation of RBS and required supporting infrastructure including a backhaul communication network
- Site, or application, specific software/firmware development and test
- Establishing and maintaining contracts for CBS services and maintenance
- Establishing and maintaining contracts for BCN services
- Developing training materials
- Development and maintenance of a Configuration Management Plan
- Documentation – operations and maintenance of hardware and software/firmware
- Maintaining a reliable database and developing a Reliability Improvement Program
- Technology insertion/migration
- Regulatory staff to ensure interoperable standards are maintained and operating procedures are enforced

Beyond cost, a deploying agency needs to consider the following factors:

- Project definition, planning, and coordination
- Funding
- Radio broadcasting licensing agreements with the Federal Communications Commission (FCC)
- Installation and maintenance

6.5 Railroads

The possible impacts for the railroad industry are:

- Interface to track-circuit based, or other, train detection system is required.

6.6 Impacts During Development and Deployment

Impacts associated specifically with development/deployment include, but may not be limited to, the following:

- Requirements for system documentation reviews and updates to as-built system
- Operational test and evaluation – possibly requiring the participation of the railroads
- Coordination with state and local agencies and the railroads
- Temporary road closures during testing
- Funding support to state and local agencies during testing
- Field changes to accommodate site-specific issues/concerns
- Issues related to technology migration

6.7 Organizational Impacts

Generally, the highway safety departments¹⁷ will ultimately assume the responsibilities associated with the RBS. The responsibilities associated with the RBS include:

- Replenishment of spares
- Training – operations and maintenance
- Preventive/scheduled maintenance¹⁸ – to include certification
- Corrective maintenance
- Ensure interoperable standards are maintained and operating procedures are enforced

The above responsibilities will most likely require additional staff.

6.8 Disadvantages and Limitations

Deployments being phased in over several years will likely be subject to changes in administrations and their respective policies and, as a result, are potentially confronted with significant impediments to wide-scale system deployment. Deploying a system, expected to be phased in over many years, will require unwavering commitment and leadership among the principal stakeholders to produce memorandums of understanding that define the roles and responsibilities associated with deploying, operating, and maintaining the system and its components. Accordingly, it is essential that long-term agreements regarding scope and funding be addressed at the outset and then periodically during the planning, design, and deployment of the system.

It is recommended that a steering committee comprised of representatives from the principal stakeholders be established to provide unambiguous direction to the project teams responsible for deploying the systems.

The willingness of the railroads to participate is a potentially limiting factor and will be critical to a successful introduction of any system concept. Critical to gaining acceptance by the railroads will be minimizing the number and significant implications of the impacts. Scenarios imposing requirements that are few in number and have minimal implications are more likely to be willingly accepted by the railroads than those that do not.

¹⁷ How the system is defined (traffic control versus HRI protection) will determine the agent responsible for its maintenance. If defined as traffic control - highway safety departments, if defined as HRI protection – railroads.

¹⁸ Maintenance-related costs are reimbursed according to formulas established by Federal and State Government agreements.

The HRI equipped with a track-circuit based train detection system scenario is one example of an approach that the railroads would be more likely to embrace. A change in state of the discrete signal that is used to control bells, flashing lights, wigwags, or gates could also be used by the RBS to initiate a change in its SPaT message. This minimalist approach does not provide the time the HRI will be occupied or recommend alternate routing to emergency vehicles. It does, however, provide the opportunity to demonstrate the efficacy of the other HRI safety applications with minimal risk, cost, and impact upon the railroads.

Abbreviations and Acronyms

AAR	American Association of Railroads
AAMVA	American Association of Motor Vehicle Administrators
AASHTO	American Association of State Highway and Transportation Officials
API	Application Programming Interface
AREMA	American Railway Engineering and Maintenance-of-Way Association
BCN	Backhaul Communications Network
CBS	cloud-based subsystem
CFR	Code of Federal Regulations
CICAS	Cooperative Intersection Collision Avoidance Systems
ConOps	Concept of Operations
CORS	Continuously Operating Reference Station
CP	Computing Platform
CV	Connected Vehicle
C-V2X	Cellular vehicle-to-everything
DOT	Department of Transportation
DC	direct current
DVI	Driver-Vehicle Interface
DSRC	dedicated short-range communication
FCC	Federal Communications Commission
FHWA	Federal Highway Administration
FRA	Federal Railroad Administration
GNSS	Global Navigation Satellite System
HRI	Highway Rail Intersection
IEEE	Institute of Electrical and Electronics Engineers
ITE	Institute of Transportation Engineers
ITS	Intelligent Transportation System
MAP	Map Data
MUTCD	Manual of Uniform Traffic Control Devices
NEMA	National Electrical Manufacturers Association
NHTSA	National Highway Safety Administration
NTRIP	Network Transport of RTCM via Internet Protocol
VBS	Vehicle-based Subsystem
OEM	Original Equipment Manufacturer
RBS	road-based subsystem
RCVW	Rail Crossing Violation Warning
RTCM	Radio Technical Commission for Maritime Services
SaaS	Software-as-a-Service
SAE	Society of Automotive Engineers
SPaT	Signal Phase and Timing
SRS	System Requirements Specification

TCP/IP	Transmission Control Protocol/Internet Protocol
TMC	Traffic Management Centers
US DOT	United States Department of Transportation
V2X	Vehicle-to-Everything
VBS	Vehicle based subsystem
VMS	Variable Message Sign
WAN	Wide-Area Network
WAVE	Wireless Access in the Vehicular Environment

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